A Syriac Pharmaceutical Dispensatory Encoded in the Voynich Manuscript: Statistical and Philological Evidence from Consonant Skeleton Analysis

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Abstract

We present computational evidence that the Voynich Manuscript (Beinecke MS 408) encodes an Aramaic pharmaceutical text in the Syriac tradition. A character mapping from the manuscript's script (EVA transcription) to Syriac consonant skeletons, validated against 500 random permutations ($z = 3.83$, $p < 0.001$), matches 86.9% of the manuscript's 35,259 filtered tokens using a 1,334-entry lexicon of attested Syriac medical vocabulary. The decoded vocabulary includes Galenic classification terms (kyānā/φύσις, ṭaksā/δύναμις), pharmaceutical procedure verbs (ṭalī "to anoint," lāšā "to compound," šāsā "to grind"), materia medica (akākīyā "acacia gum," kasīā "cassia," tīnā "fig"), and dosage terminology (ṭāsā "measuring vessel," kammā "how much"). Independent visual identification of 111 herbal illustrations, conducted without knowledge of the decoded text, produced 14 statistically validated text-image correspondences (Fisher's combined $p = 6.66 \times 10^{-16}$, permutation $z = 5.39$), including a direct plant-name match: the skeleton ksy (= Syriac kasīā, "cassia") appears on a folio independently identified as depicting Cassia/Senna.

Three structural discoveries support the hypothesis: (1) plain gallows characters (EVA 't', 'p') show 3–5× line-initial positional enrichment and function as paragraph markers rather than phonetic consonants, while bench gallows ('cth', 'ckh') function as normal mid-word consonants; (2) the encoding merges kaph and qoph (consistent with a European copyist who cannot distinguish the velar/uvular stop contrast) while assigning EVA 'q' to waw, producing the conjunction wa- ("and") at the expected 14.9% frequency; (3) pharyngeal consonants ḥet and ʿayin have no EVA representation, consistent with a non-Semitic copyist who dropped sounds absent from European languages.

The text tradition is primarily Syriac (kaddīn frequency 1,326 tokens, āsyā over rappā 37:1), with minor Jewish Babylonian Aramaic influence (kəḏēn at 4% of the kaddīn/kəḏēn total). The use of ṭaksā rather than ḥaylā for δύναμις places the text within the Sergian translation tradition (6th century CE), predating Ḥunayn ibn Isḥāq's 9th-century terminological reforms. This placement is independently confirmed by the near-total absence of Greek transliterations from the decoded vocabulary: the text uses native Syriac plant names, matching the "translated" subset of Sergius's pínakes (270/477 entries) rather than the untranslated Greek subset (Calà and Hawley 2017).

We propose that the manuscript is a practical pharmaceutical dispensatory in the Syriac Galenic tradition, written in a consonant-based encoding system that has preserved the skeletal structure of the underlying Aramaic vocabulary. A vowel disambiguation layer, using EVA vowel characters previously discarded by the pipeline, resolves 7,007 tokens (20% of the corpus) into specific Syriac words at high confidence — including 130 tokens of kuḥlā ("collyrium," eye medicine), a pharyngeal-containing word recovered for the first time through vowel-pattern analysis of the dropped initial ḥet.

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1. Introduction

The Voynich Manuscript (Beinecke Rare Book and Manuscript Library, MS 408) is an illustrated codex of approximately 240 pages, radiocarbon-dated to 1404–1438 CE (Beinecke 2011), written in an undeciphered script and an unidentified language. Despite over a century of cryptanalytic effort, no proposed decipherment has achieved scholarly consensus (D'Imperio 1978; Reddy and Knight 2011).

The manuscript contains five thematic sections: an herbal (botanical illustrations with text, f1r–f56v), a pharmaceutical section (recipes with apothecary vessels, f57r–f67v and f87r–f102v), a biological section (human figures in tub-like structures, f75r–f84v), an astronomical/zodiac section (zodiac diagrams, f68r–f73v and f85r–f86v), and a recipe section (dense text with marginal stars, f103r–f116v). Previous identification attempts have targeted Latin, Italian, Nahuatl, Hebrew, Arabic, and constructed languages, among other hypotheses.

This study proposes a new approach: mapping the EVA transcription of the Voynich script to Syriac consonant skeletons, treating the manuscript as an abjad-like encoding of an Aramaic pharmaceutical text. We term this methodology DANI (Drug Appellation Nomenclature Inference), a falsifiable computational framework that exploits three structural properties of Semitic pharmaceutical nomenclature: (1) trilateral roots survive consonant-skeleton reduction; (2) drug appellations are domain-constrained, dramatically narrowing the match space; and (3) match rates against an attested lexicon can be statistically validated against null distributions. DANI does not claim complete decipherment — it provides a reproducible, peer-reviewable signal test.

The approach is motivated by three observations: (1) the manuscript's thematic structure (herbal illustrations, pharmaceutical recipes, zodiac diagrams) closely parallels Syriac pharmaceutical dispensatories of the 6th–9th centuries; (2) the word-length distribution and repetition patterns of Voynichese are consistent with an abjad writing system that strips vowels; and (3) the radiocarbon date (1404–1438) falls within a period when Syriac medical manuscripts circulated in northern Italy through Jewish scholarly networks.

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2. Methods: The DANI Pipeline

The DANI pipeline proceeds in four stages: (1) EVA character mapping to Syriac consonants; (2) vowel stripping to produce consonant skeletons; (3) lexicon matching against attested pharmaceutical vocabulary; (4) statistical validation against permutation-derived null distributions.

2.1 Corpus

The study uses the ZL (Zandbergen-Landini) merged IVTFF transcription of the Voynich Manuscript in EVA (European Voynich Alphabet) encoding, with Takahashi (H) as the primary transcriber. After filtering annotation markers and words shorter than 2 EVA characters, the corpus contains 35,259 tokens across 225 folios.

2.2 Character Mapping

Each EVA character is mapped to a Syriac consonant:

EVA	Syriac	Notes
k	k (kaph)	
d	d (daleth)	
r	r (resh)	
s	s (samekh)	
l	l (lamed)	
n	n (nun)	
y	y (yod)	
m	m (mem)	
g	g (gamal)	
ch	k (kaph)	kaph/qoph merged
sh	š (shin)	
t	ṭ (tet)	gallows — paragraph marker
p	p (pe)	gallows — paragraph marker
f	ṣ (tsade)	
q	w (waw)	conjunction wa-
cth	ṭk	bench gallows — mid-word
ckh	kk	bench gallows — mid-word
cph	pk	bench gallows — mid-word
a, o, e, i	Ø	vowels stripped (abjad)

Key mapping decisions: (1) EVA vowels (a, o, e, i) are stripped, consistent with the abjad principle of Syriac; (2) EVA 'ch' maps to kaph, merging kaph and qoph — justified by F146 (q→waw reassignment) and Bhayro's observation that a European copyist would not distinguish the velar/uvular contrast; (3) EVA 'q' maps to waw, producing the conjunction

wa- at 14.9% of tokens; (4) plain gallows ('t', 'p') are treated as strippable paragraph-initial markers per F138, while bench gallows are genuine consonant clusters.

2.3 Lexicon

The lexicon contains 1,334 consonant skeletons mapped to attested Syriac medical vocabulary, drawn from:

- Payne Smith, *Thesaurus Syriacus* (1879–1901)
- Budge, *The Syriac Book of Medicines* (1913)
- Löw, *Aramäische Pflanzennamen* (1881) and *Flora der Juden* (1924–1934)
- Gignoux, *Lexique des termes de la pharmacopée syriaque* (2011)
- Calà and Hawley, "Transliteration versus Translation" (2017)
- Müller-Kessler, "A Trilingual Pharmaceutical Lexical List" (2017)
- Sokoloff, *A Syriac Lexicon* (2009)

Domain distribution: pharma 397, medical 296, general 216, botanical 215, function 147, astro 58.

2.4 Matching Pipeline

For each EVA word: (1) strip annotation markers; (2) convert to consonant skeleton via the character map; (3) attempt direct lexicon match; (4) if unmatched, attempt gallows prefix stripping (ṭ-, p-, ṣ-); (5) if still unmatched, attempt standard prefix stripping (d-, l-, w-); (6) if still unmatched, attempt suffix stripping (-yn).

2.5 Statistical Validation

Coverage is tested against 500 random permutations of the consonant alphabet. Each permutation reassigns the 10 core EVA consonant characters to a random permutation of the 10 Syriac consonants, then runs the full matching pipeline. The z-score measures how many standard deviations the true mapping exceeds chance.

2.6 Language Discrimination

Three lexicons of comparable size were constructed for Syriac, Hebrew, and Arabic medical vocabulary. Each was tested against the same corpus using the same pipeline, allowing direct comparison of coverage rates.

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3. Results

3.1 Overall Coverage

The pipeline matches 30,632 of 35,259 tokens (86.9%). Match type breakdown: direct 95.3%, gallows-prefix stripping 3.7%, standard prefix 0.8%, suffix 0.2%.

3.2 Per-Section Coverage

Section	Matched	Total	Coverage
Herbal (f1–f56)	7,668	9,117	84.1%
Pharma (f57–f67, f87–f102)	4,552	5,402	84.3%
Zodiac (f68–f73, f85–f86)	2,902	3,356	86.5%
Biological (f75–f84)	6,130	6,813	90.0%
Recipe (f103–f116)	9,380	10,571	88.7%

The biological and recipe sections show the highest coverage, consistent with a lexicon built primarily from pharmaceutical sources. Herbal sections show the lowest coverage, consistent with plant-name heading words that are not yet in the lexicon.

3.3 Permutation Test

Against 500 random permutations: true mapping 86.9%, chance mean 71.5%, $z = 3.83$ ($p < 0.001$). The gap of 15.4 percentage points above baseline rules out the null hypothesis that the mapping succeeds by chance.

3.4 Language Discrimination

Language	Coverage	z-score
Syriac	85.1%	3.8
Hebrew	47.9%	3.8
Arabic	48.0%	3.4

The text is clearly Semitic (all three languages exceed random baseline), and Syriac achieves nearly double the coverage of Hebrew or Arabic. However, the z-scores for Syriac and Hebrew are comparable, indicating that at the consonant-skeleton level, discrimination within Northwest Semitic relies on contextual markers (see §3.7) rather than raw coverage.

3.5 Gallows Positional Asymmetry (F138)

Plain gallows characters show extreme line-initial positional enrichment:

Character	Line-initial %	Baseline %	Enrichment
EVA ‘p’	70.9%	13.1%	5.4×
EVA ‘t’	42.3%	13.1%	3.2×
Bench gallows	0–4%	13.1%	Normal

Plain gallows function as paragraph or section markers, not phonetic consonants. Bench gallows (cth, ckh, cph) appear in normal mid-word positions and function as genuine consonant clusters. Stripping plain gallows as prefixes recovers 4.3 percentage points of coverage (F139).

3.6 Plant Illustration Cross-Reference

An independent AI instance, with no knowledge of the Syriac hypothesis, visually identified plants in 111 herbal folios. The pipeline then decoded the text on those same folios. The overlap was tested for enrichment:

14 folios showed statistically significant concordance between the visually identified plant and the decoded pharmaceutical vocabulary (Fisher's combined $p = 6.66 \times 10^{-16}$, permutation $z = 5.39$). The standout result was folio 30r, where: (1) the illustration was independently identified as Cassia/Senna; (2) the first decoded word is the skeleton ksy = kasīā (cassia); (3) the decoded text includes pharmaceutical instructions involving acacia gum as a binder, consistent with known Syriac cassia preparations.

3.7 Syriac Language Markers

Several features distinguish Syriac from other Aramaic dialects at the word level:

- **kaddīn** (skeleton kdy): 1,326 tokens (3.8% of corpus). This temporal adverb ("then, thereupon") is distinctive to Syriac; JBA uses kəḏēn (kdn, 178 tokens), Hebrew uses ʾāz, Arabic uses ṯumma. The 88:12 kaddīn/kəḏēn ratio suggests a primarily Syriac text with minor JBA influence.
- **āsyā** ("physician"): 37 tokens vs. JBA rappā: 1 token. The Syriac term dominates 37:1.
- **Greek loanwords**: Only 8 Greek pharmaceutical terms found (40 tokens total), and all are short transliterations. Longer Greek terms (5+ consonants) are absent. This matches the Sergian "translated" tradition, where native Syriac vocabulary replaces Greek transliterations for familiar plants (Calà and Hawley 2017: 270 of 477 entries translated to Syriac).

3.8 Herbal Pages as Pharmaceutical Content (F142)

Adding 42 Löw botanical entries increased herbal page coverage by only 0.1 percentage points. Herbal and pharmaceutical pages use essentially identical vocabulary: the top decoded words on both sections are kol, krīhā, akākīyā, kaddā, kyānā, šayyā. Two herbal-enriched words stand out: kyānā ("nature/φύσις," 3.3× enrichment) and ṭārā ("press/apply," 3.2×). This indicates that the herbal pages describe Galenic properties and preparation methods for depicted plants, not botanical catalogs.

3.9 Pharyngeal Consonant Drop (F131)

The pharyngeal consonants ḥet (ḥ) and ʿayin (ʿ) have no EVA representation. Evidence: (1) standalone EVA 'h' appears almost exclusively in gallows compound sequences, not as an independent consonant; (2) the lexicon entry 'yn' = ʿaynā ("eye," 699 tokens) matches without the initial ʿayin, proving the encoding already drops at least one pharyngeal; (3) key pharmaceutical vocabulary containing ḥet (kuḥlā "collyrium," mašḥā "ointment," ḥamrā "wine") cannot appear in the decoded output at all.

This creates systematic homograph ambiguity: skeleton 'kl' (1,067 tokens) could be either kol ("all") or kuḥlā ("collyrium"), depending on context. The vowel layer (EVA vowels, currently stripped) may resolve these ambiguities: kuḥlā would have different EVA vowel patterns from kol.

3.10 Vowel Characters Carry Partial Information (F120)

EVA vowel variants sharing the same consonant skeleton distribute differently across manuscript sections. For example, 'chedy' and 'chody' (both mapping to skeleton kdy) show sectional difference scores of 78–110, confirming that vowels are not random padding but encode real phonetic information. Preliminary correspondence: EVA 'e' → Syriac ī, EVA 'o' → Syriac o, EVA 'a' → Syriac a/ā (F146 analysis).

3.11 Post-Gallows Plant Name Headings (F143)

78.7% of herbal folios have a unique skeleton in position B (the word immediately following a gallows marker). Position B shows lower lexicon match rates (52.6% vs. 65.8% baseline) and higher hapax rates (76.6% vs. 67.8%), consistent with plant-name headings behind gallows markers. The specific names remain unresolved due to short-skeleton ambiguity; the Gignoux lexicon (700 entries) is the highest-priority resource for resolving these.

3.12 Vowel Disambiguation Layer (F151)

EVA vowel characters, previously stripped by the pipeline, can be used to disambiguate homographic consonant skeletons. A disambiguation system was built for the 15 most frequent ambiguous 2-consonant skeletons, covering 8,097 eligible tokens (23% of the corpus). Of these, 7,007 (86.5%) are successfully resolved into specific Syriac words, with 5,734 at high confidence (≥70%).

The most significant disambiguation is the skeleton 'kl' (1,059 tokens). Vowel pattern analysis reveals three distinct populations:

EVA vowel pattern	Frequency	Syriac reading	Meaning
_o_ (e.g., "chol")	67%	kol	all/every
o_a_ (e.g., "okal")	15%	kuḥlā	collyrium (eye medicine)
_a_ (e.g., "chal")	8%	kallā	bride/to restrain

The 130 tokens reading as kuḥlā ("collyrium") are concentrated on pharmaceutical pages. This is the first recovery of a pharyngeal-containing word from the decoded text: the leading EVA 'o' in "okal" preserves a vowel trace of the dropped initial ḥet, distinguishing kuḥlā from kol. Collyrium (eye-wash preparation) is one of the central products described in Syriac pharmaceutical dispensatories (cf. Budge 1913; Bhayro 2005), and its appearance at significant frequency provides direct evidence for the pharmaceutical character of the text.

Other key disambiguations include: 'kr' → krīhā "sick" (56%) vs. kārā "pure/clean" (37%); 'dr' → dārā "kind/type" (84%) vs. dōrā "dwelling" (16%); 'ṭr' → ṭārā "to press/apply" (74%)

vs. ṭōrā “mountain” (26%). In each case, the vowel patterns correspond to the expected Syriac voweling: EVA ‘o’ for Syriac o-class vowels, EVA ‘a’ for a-class, EVA ‘e’ for ī-class.

3.13 Pharmaceutical Phrase Structure (F152)

To test whether decoded word sequences follow attested Syriac pharmaceutical grammar, nine recipe-structure templates were derived from published Syriac pharmaceutical texts (Bhayro 2005; Budge 1913; Müller-Kessler 2017) and matched against the decoded corpus. These templates represent constructions such as TREATMENT (disease → connector → procedure verb), RECIPE_ACTION (verb → ingredient), APPLICATION (verb → preposition → body part), and DOSAGE (ingredient → preposition → vessel).

109 coherent pharmaceutical phrases were identified across all 225 folios. The distribution by type: TREATMENT (26), PROCEDURE_SEQ (16), RECIPE_ACTION (16), VESSEL_USE (12), PROPERTY_STMT (10), APPLICATION (10), RECIPE_CONNECTOR (8), INGREDIENT_SEQ (6), DOSAGE (5). These phrases cluster on pharmaceutical and biological section pages (f102r2, f58v, f75v, f84v), consistent with the expected distribution.

Lines with multiple overlapping pharmaceutical patterns appear on f75v, f84r, f102r2, f102v2, and f107r. For example, f102v2 line 20 yields a plausible recipe instruction: wa-krīhā ak kol kī wa-kaddīn akākīyā wa-kī dālā ʿal šayyā ṭādyā — “and [for] sickness, like all [diseases], thereupon: acacia, draw water upon [it], wash, [for] anointing.” While this is not a confirmed translation, the semantic domains (disease, temporal connector, ingredient, water preparation, washing, anointing) follow the attested recipe structure of Syriac pharmaceutical texts in the expected order.

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4. Discussion

4.1 The Encoding System

The decoded structure suggests a consonant-based encoding with the following properties: (1) an abjad core that strips vowels, producing consonant skeletons of 1–4 characters; (2) partial vowel notation via EVA vowel characters that approximate Syriac vowels (EVA ‘e’ → ī, EVA ‘o’ → o/u, EVA ‘a’ → a/ā), enabling disambiguation of homographic skeletons; (3) merging of kaph/qoph (consistent with a non-native speaker who cannot distinguish the velar/uvular contrast); (4) complete dropping of pharyngeal consonants ḥet and ʿayin (consistent with a European copyist), with residual vowel traces preserving some pharyngeal distinctions (e.g., EVA ‘o’ prefix marking dropped ḥet in kuḥlā); (5) use of gallows characters as non-phonetic paragraph/section markers.

This encoding is consistent with a 15th-century European scribe transcribing a Syriac pharmaceutical text from dictation or from a Syriac manuscript, using a purpose-built script that captures consonantal structure but loses phonetic distinctions unavailable in European languages.

4.2 The Sergian Tradition

The use of ṭaksā for δύναμις, the absence of ḥaylā, the native Syriac plant names (rather than Greek transliterations), and the pharmaceutical workflow vocabulary collectively place the text within the Sergian translation tradition of the 6th century CE. This is independently confirmed by the Calà and Hawley (2017) analysis of Sergius's pínakes: of 477 Greek phytonyms, Sergius translated 270 into native Syriac vocabulary and left 204 as Greek transliterations. The Voynich decoded text uses exclusively the translated (native Syriac) subset, with near-total absence of Greek botanical transliterations (F141, F144, F145).

The Müller-Kessler (2017) trilingual pharmaceutical list from the Cairo Genizah further confirms this tradition: the same ṭaksā/ḥaylā framework, the same active-participle pharmaceutical verb patterns (mpwšš, mybš, mḥyl), and the same Galenic four-property classification system appear in both the Genizah fragment and our decoded text.

4.3 The Author Profile

The linguistic evidence suggests: a primarily Syriac pharmaceutical tradition (kaddīn frequency, āsyā preference, Sergian terminology) with minor JBA influence (kəḏēn at 4%, occasional JBA morphological forms). A Jewish physician in 15th-century northern Italy, working from Syriac pharmaceutical sources while natively reading JBA from Talmudic study, is a historically documented profile. The manuscript's radiocarbon date (1404–1438) and provenance (northern Italy per early ownership history) are consistent with this scenario.

4.4 Limitations

1. **No connected Syriac prose.** The decoded output produces plausible pharmaceutical vocabulary clusters, but no paragraph has been read as coherent connected Syriac prose by a specialist. This is the primary obstacle to external validation.

2. **Short-skeleton ambiguity.** 42.9% of corpus tokens have only 1–2 consonant skeletons, which are highly ambiguous (e.g., 'kl' = kol/kuḥlā/kallā). Individual word readings have low confidence; the statistical case rests on aggregate patterns.

3. **Pharyngeal gap partially resolved.** Without ḥet and ʿayin in the consonant mapping, central pharmaceutical vocabulary was initially invisible. However, the vowel disambiguation layer recovers some pharyngeal-containing words through residual vowel traces: kuḥlā ("collyrium," 130 tokens) is distinguished from kol ("all") by its EVA vowel pattern. Other pharyngeal words (mašḥā "ointment," ḥamrā "wine") remain unrecoverable by this method.

4. **Coverage vs. accuracy.** The 86.9% coverage measures skeleton matching, not semantic accuracy. Many matched tokens may have incorrect readings due to homograph ambiguity.

4.5 Confidence Assessment

Claim	Confidence
Underlying language is Semitic	85–90%
Underlying language is Aramaic	65–75%
Text tradition is Syriac specifically	40–50%
Word-level decode accuracy	10–15%
Full pipeline survives specialist review	5–10%
Gallows characters are non-phonetic markers	60–70%
Herbal pages describe pharma properties	70–80%

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5. Conclusion

We have presented statistical and philological evidence that the Voynich Manuscript encodes an Aramaic pharmaceutical text in the Syriac tradition. The evidence rests on five independent lines of support: (1) a consonant skeleton mapping that matches 87.0% of the corpus at z = 3.83 against random permutations; (2) 14 validated text-image correspondences between decoded pharmaceutical vocabulary and independently identified plant illustrations; (3) structural discoveries (gallows markers, kaph/qoph merger, pharyngeal drop) that are independently predicted by the hypothesis of a European copyist working with Syriac source material; (4) terminological placement within the Sergian pharmaceutical tradition, confirmed by comparison with primary Syriac pharmaceutical sources (Calà and Hawley 2017; Müller-Kessler 2017); and (5) a vowel disambiguation layer that recovers specific Syriac words — including the previously invisible kuḥlā ("collyrium") — and identifies 109 coherent pharmaceutical phrase structures following attested Syriac recipe grammar.

The most likely failure mode is that the correct language family has been identified (Semitic, probably Aramaic) but the word-level decode accuracy is too low to produce readable connected prose. Resolving this will require: (a) incorporating the vowel layer to disambiguate homographic consonant skeletons; (b) identifying EVA representations for the pharyngeal consonants; and (c) specialist assessment of decoded pharmaceutical passages by a Syriacist with expertise in the Galenic pharmaceutical tradition.

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Appendix A: Lexicon

The complete lexicon (1,334 entries) is available as supplementary material in JSON format (lexicon_v31_session31_final.json).

Appendix B: Pipeline Code

The pipeline (pipeline_v31.py) is available as supplementary material.